

Module 2: Overview

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Module Highlights

In Part 1 of this module, you will learn how to compare means in two situations: (1) two independent samples, and (2) paired samples. We will focus on the ideas behind confidence intervals and p -values, which are based on the t -distribution. These methods are developed under the assumption of random sampling from normally distributed populations. In Part 2 you will learn about distribution-free methods for comparing population means, based on models that do not specify any particular population distributions, for situations where the t -tools assumptions are grossly violated. In Part 3 of this module, you will learn how to summarize proportions with contingency tables, marginal and conditional probabilities, how to compare two proportions using differences and odds ratios, and how to apply several inferential tools derived specifically for comparing proportions.



This module is designed to be studied during a span of two and a half weeks. You should plan to complete this module by the end of week 5.

Module Learning Objectives (MLOs)

Upon completion of this module, you should be able to:

- **MLO1: Apply and interpret t-tests** by comparing randomization and t -distributions and constructing confidence intervals for differences in means.
- **MLO2: Recognize and analyze paired data** by identifying when a paired approach is appropriate, applying correct methods, and interpreting results.
- **MLO3: Explain the differences between frequentist and Bayesian perspectives**, and interpret confidence intervals in relation to credible intervals.
- **MLO4: Use nonparametric alternatives to the t-test**, including the Wilcoxon rank-sum, sign test, and Wilcoxon signed-rank test, and explain when these methods are useful.
- **MLO5: Compare group proportions** using contingency tables, marginal and conditional probabilities, and mosaic plots.
- **MLO6: Estimate and perform inference** for the difference in two proportions.
- **MLO7: Estimate and interpret odds ratios** and perform inference for odds ratios in the context of 2×2 contingency tables.

- **MLO8: Conduct and interpret chi-square and McNemar's tests** to analyze categorical data in independent and paired-sample settings.

Activities & Assignments

In this module, you will:

- **Read** through the Lecture Notes and, optionally, Chapters 2 and 18 in the Statistical Sleuth (recommended but not required text).
- **Watch** the Lecture Videos.
- **Do** the Lab 03 and Lab 04 Activities
- **Work** through the code yourself in R
- **Submit** HW03 and HW04